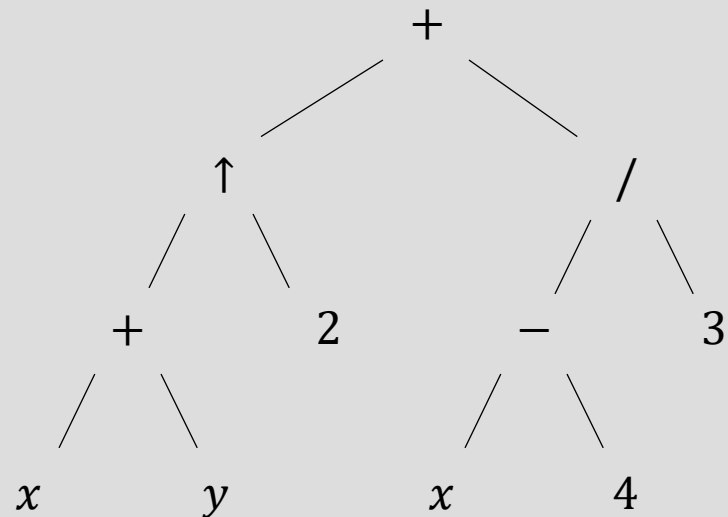


TREES

CSI 219 – Discrete Mathematics

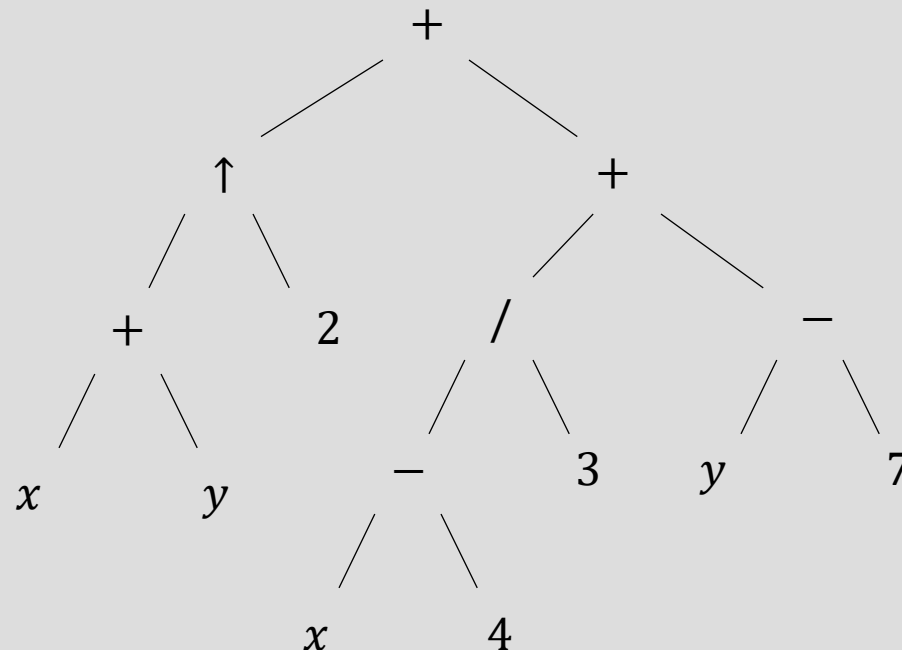
ORDERED ROOTED TREE OF EXPRESSIONS

$$((x + y) \uparrow 2) + ((x - 4)/3)$$

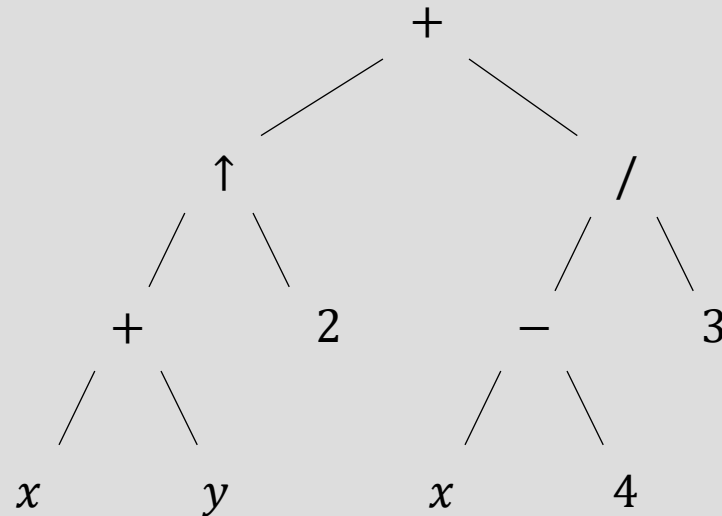


ORDERED ROOTED TREE OF EXPRESSIONS

$$((x + y) \uparrow 2) + (((x - 4)/3) + (y - 7))$$



TRAVERSAL OF THE TREE: INFIX, PREFIX AND POSTFIX NOTATIONS



- In-order: $x + y \uparrow 2 + x - 4 / 3 \rightarrow$ Infix notation
- Pre-order: $+ \uparrow + x y 2 / - x 4 3 \rightarrow$ Prefix notation
- Post-order: $x y + 2 \uparrow x 4 - 3 / + \rightarrow$ Postfix notation